

Firewall acceptance matrix

Test	Source	Destination / service	Expected	Reason
FW-01	Office client	Business application HTTPS	PASS	Normal business access
FW-02	Office client	Backup service/API	BLOCK	Office cannot reach backup repository
FW-03	Office client	Management VLAN 99	BLOCK	Normal users cannot reach administration
FW-04	Office endpoint	Monitoring TCP 1514	PASS if agent deployed	Approved telemetry path
FW-05	Office endpoint	Monitoring TCP 1515	PASS only when enrolment required	Approved enrolment path
FW-06	APP01	BKP01 prototype TCP 9000	PASS	Validate push backup path
FW-07	APP01	MON01 prototype TCP 1514	PASS	Validate server telemetry path
FW-08	Office client	Firewall management TCP 443	BLOCK	Users cannot manage firewall
FW-09	ADMIN-PC	Firewall management TCP 443	PASS	Approved management path
FW-10	ADMIN-PC	APP01 TCP 22/443	PASS	Approved server/application administration
FW-11	ADMIN-PC	MON01 TCP 22/443	PASS	Approved monitoring administration
FW-12	ADMIN-PC	BKP01 admin/API ports	PASS	Approved backup administration
FW-13	BKP01 / Backup zone	APP01 / Server zone unsolicited connection	BLOCK	Backup zone should not freely initiate traffic into Server zone
FW-14	Internal host	Approved DNS resolver UDP/TCP 53	PASS	Required name resolution
FW-15	Internal host	Approved NTP source UDP 123	PASS	Required time synchronisation

Executed prototype firewall test matrix

The table below shows the 15 firewall tests carried out on the three active prototype systems. Expected service paths to connect required service and unapproved service paths to time out. For the screen shot refer to Appendix A.

Test	Source	Destination / service	Expected	Observed result	Outcome	Evidence
FT-01	APP01 10.20.20.10	MON01 10.20.30.10 TCP 1514	ALLOW	Connection succeeded	PASS	Figure A1
FT-02	APP01 10.20.20.10	MON01 10.20.30.10 TCP 1515	ALLOW	Connection succeeded	PASS	Figure A2
FT-03	APP01 10.20.20.10	BKP01 10.20.40.10 TCP 9000	ALLOW	Connection succeeded	PASS	Figure A3
FT-04	BKP01 10.20.40.10	MON01 10.20.30.10 TCP 1514	ALLOW	Connection succeeded	PASS	Figure A4
FT-05	BKP01 10.20.40.10	MON01 10.20.30.10	ALLOW	Connection succeeded	PASS	Figure A5

Test	Source	Destination / service	Expected	Observed result	Outcome	Evidence
		TCP 1515				
FT-06	APP01 10.20.20.10	BKP01 10.20.40.10 TCP 22	BLOCK	Connection timed out	PASS	Figure A6
FT-07	MON01 10.20.30.10	BKP01 10.20.40.10 TCP 9000	BLOCK	Connection timed out	PASS	Figure A7
FT-08	BKP01 10.20.40.10	APP01 10.20.20.10 TCP 443	BLOCK	Connection timed out	PASS	Figure A8
FT-09	MON01 10.20.30.10	APP01 10.20.20.10 TCP 22	BLOCK	Connection timed out	PASS	Figure A9
FT-10	MON01 10.20.30.10	BKP01 10.20.40.10 TCP 22	BLOCK	Connection timed out	PASS	Figure A10
FT-11	BKP01 10.20.40.10	MON01 10.20.30.10 TCP 22	BLOCK	Connection timed out	PASS	Figure A11
FT-12	BKP01 10.20.40.10	APP01 10.20.20.10 TCP 22	BLOCK	Connection timed out	PASS	Figure A12
FT-13	APP01 10.20.20.10	BKP01 10.20.40.10 TCP 9001	BLOCK	Connection timed out	PASS	Figure A13
FT-14	APP01 10.20.20.10	MON01 10.20.30.10 TCP 22	BLOCK	Connection timed out	PASS	Figure A14
FT-15	MON01 10.20.30.10	APP01 10.20.20.10 TCP 443	BLOCK	Connection timed out	PASS	Figure A15

Result summary: 15 out of 15 prototype firewall tests were performed and the expected result was obtained. 10 unnecessary service paths were blocked and 5 necessary Wazuh/MinIO service paths were allowed. These results only apply to the APP01, MON01 and BKP01 flows tested; they are not a substitute for the unexecuted Office, Management, DNS or NTP acceptance tests above.

Access-control tests

Test	Control	Expected evidence
AC-01	Finance user opens Finance folder	Finance account can read/write Finance data but cannot gain Backup or firewall administration.
AC-02	Finance user attempts Sales/Operations restricted data	Access denied or not presented according to the configured role permissions.
AC-03	Sales user opens Sales folder and attempts Finance folder	Own department access succeeds; Finance access is blocked.
AC-04	Operations user opens Operations folder and attempts Finance/Sales restricted data	Own department access succeeds; unrelated department access is blocked.
AC-05	General Staff account	Shared information is available but restricted department folders are blocked.
AC-06	Management folder permissions	Management receives only the permissions documented in the final role design.
AC-07	Nextcloud administrator MFA in prototype	TOTP challenge is required and successful login is recorded.
AC-08	Normal user attempts monitoring administration	No administrative access.
AC-09	Normal user attempts backup administration	No administrative access.
AC-10	APP01 backup service identity	Backup succeeds through the dedicated path without backup-console administrative rights.
AC-11	Leaver simulation	Test account is disabled and can no longer authenticate.
AC-12	Privilege review	Privileged accounts and group memberships match the approved matrix.
AC-13	Logging	At least one controlled access or deny event is visible with source and timestamp in the expected log location.

Executed prototype access-control evidence

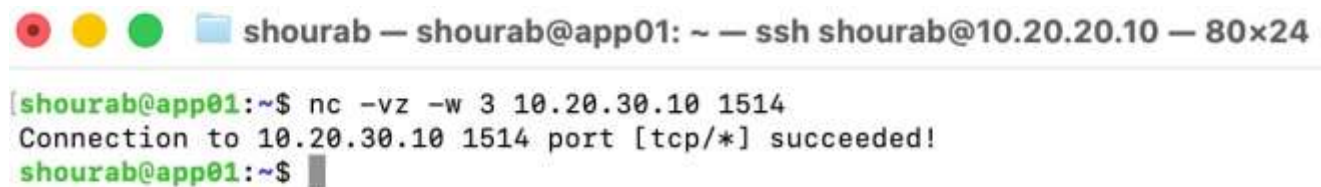
The results from the following tests document the evidence collected in the Nextcloud prototype concerning access-control. The evidence indicates that each different user is getting different folders as per their role, a Finance user will not have direct access to Sales data and administrator login needs a TOTP second factor. Appendix C contains the screenshots referred to.

Test	Control	Expected	Observed result	Outcome	Evidence
AC-E01	Finance officer - Finance folder	ALLOW / Read-Write	Finance folder opened and Test File.md is visible after the Finance user created it.	PASS	Figure C3
AC-E02	Finance officer - Sales folder direct access	BLOCK	Direct attempt to open the Sales folder returned "Folder not found".	PASS	Figure C4
AC-E03	Manager - approved business folders	ALLOW	Manager account shows Company Shared, Finance, Management, Operations and Sales folders.	PASS	Figure C1
AC-E04	Foysal prototype account - role-specific folders	ALLOW / restricted by role	IT and Backup folder is visible with Company Shared, while Finance, Management, Operations and Sales are not shown.	PASS	Figure C2
AC-E05	Nextcloud administrator - MFA	TOTP REQUIRED	After password authentication, Nextcloud requires a TOTP authentication code before login can continue.	PASS	Figure C5

Result Summary: 5 of 5 executed access-control evidence checks returned the expected result. The Finance positive and negative tests show how folders are separated by department, the Manager and Foysal screenshots show how the user can only see folders related to their role, and the TOTP challenge shows how MFA is implemented on the prototype administrator login.

Appendix A Firewall Test Evidence

The screenshots below provide the command-line evidence for FT-01 to FT-15 in the executed prototype firewall test matrix. Allowed tests show a successful TCP connection. Blocked tests show a timeout. The screenshots are kept in the appendix so the main body remains readable while the evidence remains traceable. (Evidence captured during prototype firewall verification on 29 September 2026.)



```
shourab — shourab@app01: ~ — ssh shourab@10.20.20.10 — 80x24
[shourab@app01:~$ nc -vz -w 3 10.20.30.10 1514
Connection to 10.20.30.10 1514 port [tcp/*] succeeded!
shourab@app01:~$
```

Figure A1. FT-01 - APP01 to MON01 TCP 1514 – allowed



```
shourab — shourab@app01: ~ — ssh shourab@10.20.20.10 — 80x24
[shourab@app01:~$ nc -vz -w 3 10.20.30.10 1515
Connection to 10.20.30.10 1515 port [tcp/*] succeeded!
shourab@app01:~$
```

Figure A2. FT-02 - APP01 to MON01 TCP 1515 – allowed



```
shourab — shourab@app01: ~ — ssh shourab@10.20.20.10 — 80x24
[shourab@app01:~$ nc -vz -w 3 10.20.40.10 9000
Connection to 10.20.40.10 9000 port [tcp/*] succeeded!
shourab@app01:~$
```

Figure A3. FT-03 - APP01 to BKP01 TCP 9000 - allowed

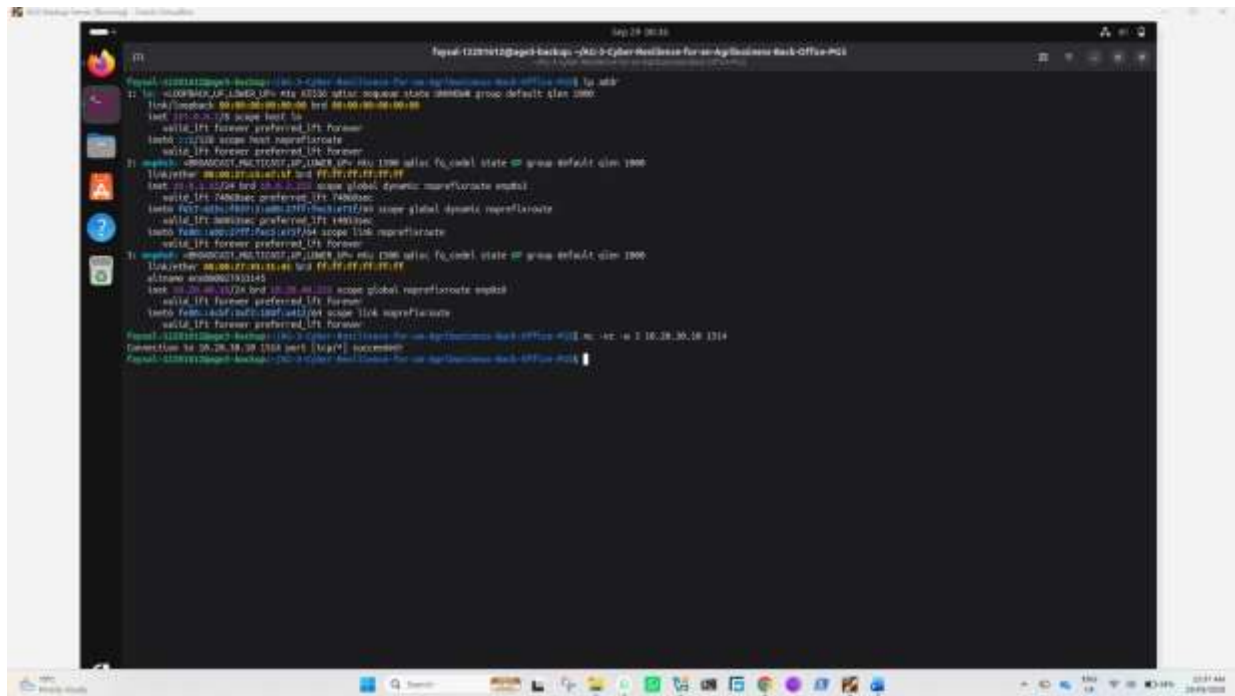


Figure A4. FT-04 - BKP01 to MON01 TCP 1514 – allowed

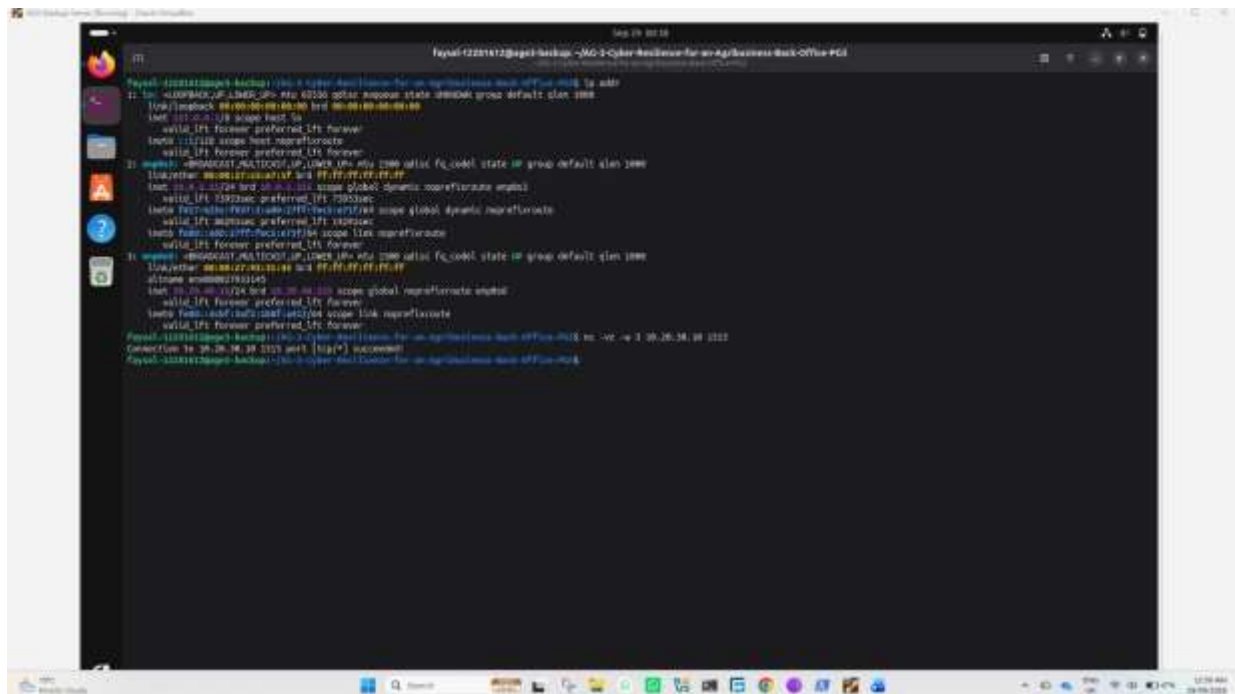


Figure A5. FT-05 - BKP01 to MON01 TCP 1515 - allowed


```

sheikh@MON01: ~
sheikh@MON01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: emp20a: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:d4:f6 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c7d4f6
    inet 172.18.200.130/24 brd 172.18.200.255 scope global dynamic noprefixroute emp20a
        valid_lft 1625sec preferred_lft 1625sec
    inet6 fe80::20c:29ff:fc7:d4f6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: emp20b: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:d4:05 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c7d405
    inet 10.20.30.10/24 brd 10.20.30.255 scope global noprefixroute emp20b
        valid_lft forever preferred_lft forever
sheikh@MON01:~$ nc -vz -w 3 10.20.30.10 22
nc: connect to 10.20.30.10 port 22 (tcp) timed out: Operation now in progress
sheikh@MON01:~$

```

Figure A9. FT-09 - MON01 to APP01 TCP 22 – blocked

```

sheikh@MON01: ~
sheikh@MON01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: emp20a: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:d4:f6 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c7d4f6
    inet 172.18.200.130/24 brd 172.18.200.255 scope global dynamic noprefixroute emp20a
        valid_lft 1522sec preferred_lft 1522sec
    inet6 fe80::20c:29ff:fc7:d4f6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: emp20b: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:d4:05 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c7d405
    inet 10.20.30.10/24 brd 10.20.30.255 scope global noprefixroute emp20b
        valid_lft forever preferred_lft forever
sheikh@MON01:~$ nc -vz -w 3 10.20.40.10 22
nc: connect to 10.20.40.10 port 22 (tcp) timed out: Operation now in progress
sheikh@MON01:~$

```

Figure A10. FT-10 - MON01 to BKP01 TCP 22 – blocked

```

sheikh@MON01: ~
sheikh@MON01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: emp20a: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:d4:f6 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c7d4f6
    inet 172.18.200.130/24 brd 172.18.200.255 scope global dynamic noprefixroute emp20a
        valid_lft 1625sec preferred_lft 1625sec
    inet6 fe80::20c:29ff:fc7:d4f6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: emp20b: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:d4:05 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c7d405
    inet 10.20.30.10/24 brd 10.20.30.255 scope global noprefixroute emp20b
        valid_lft forever preferred_lft forever
sheikh@MON01:~$ nc -vz -w 3 10.20.30.10 22
nc: connect to 10.20.30.10 port 22 (tcp) timed out: Operation now in progress
sheikh@MON01:~$

```

Figure A11. FT-11 - BKP01 to MON01 TCP 22 - blocked



Figure A12. FT-12 - BKP01 to APP01 TCP 22 – blocked



Figure A13. FT-13 - APP01 to BKP01 TCP 9001 – blocked

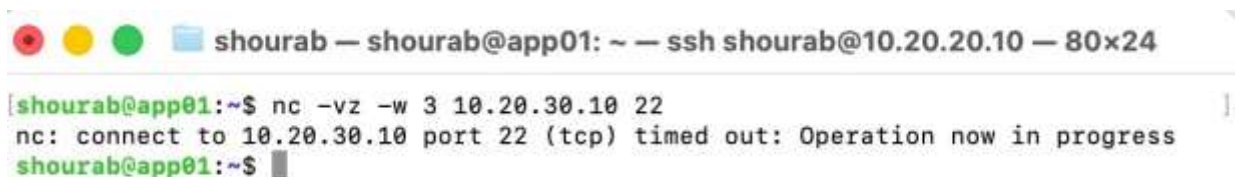


Figure A14. FT-14 - APP01 to MON01 TCP 22 - blocked

```
sheikh@MON01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens20s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:0d:05 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c70d05
    inet 172.16.200.130/24 brd 172.16.200.255 scope global dynamic noprefixroute ens20s1
        valid_lft 1210sec preferred_lft 1210sec
    inet6 fe80::20c:29ff:fc7:d005 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens20s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c7:0d:05 brd ff:ff:ff:ff:ff:ff
    altname ens00c29c70d05
    inet 10.20.20.10/24 brd 10.20.20.255 scope global noprefixroute ens20s0
        valid_lft forever preferred_lft forever
sheikh@MON01:~$ nc -v -w 5 10.20.20.10 443
nc: connect to 10.20.20.10 port 443 (tcp) timed out: Operation now in progress
sheikh@MON01:~$
```

Figure A15. FT-15 - MON01 to APP01 TCP 443 - blocked

Appendix B - pfSense Configuration Evidence

The screenshots below show the pfSense VLAN and firewall rule configuration used in the prototype. They are included as configuration evidence to support the executed connectivity tests in Appendix A.

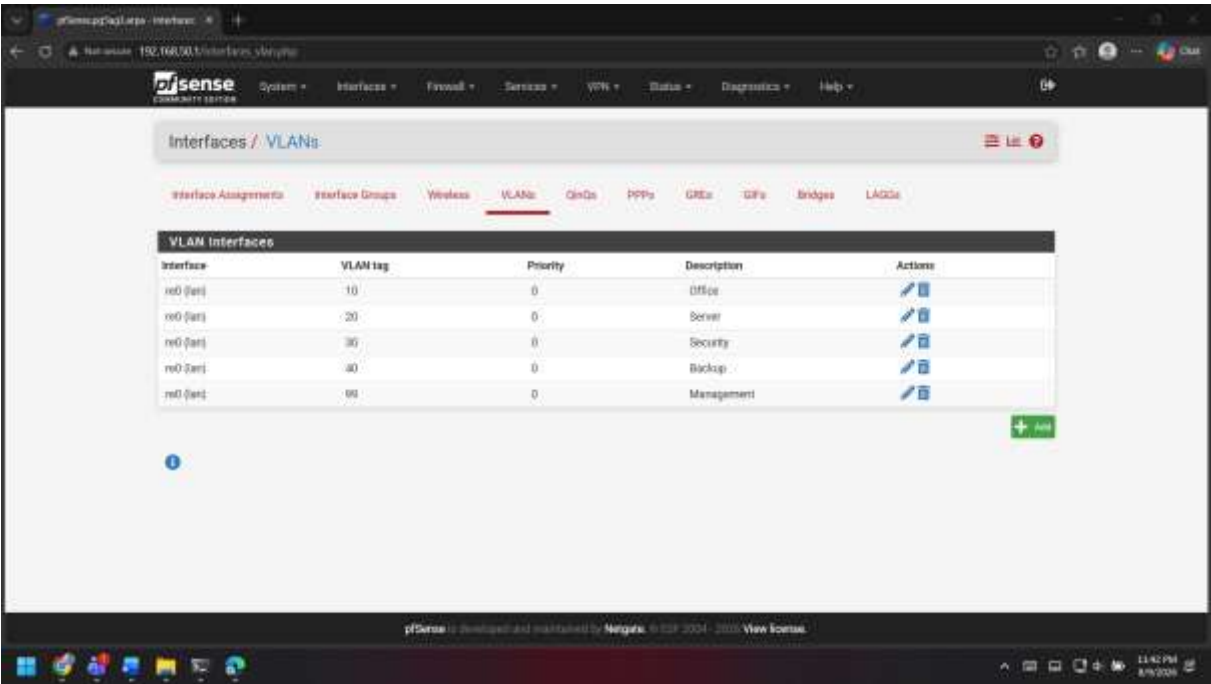


Figure B1. pfSense VLAN interfaces configured for Office, Server, Security, Backup and Management.

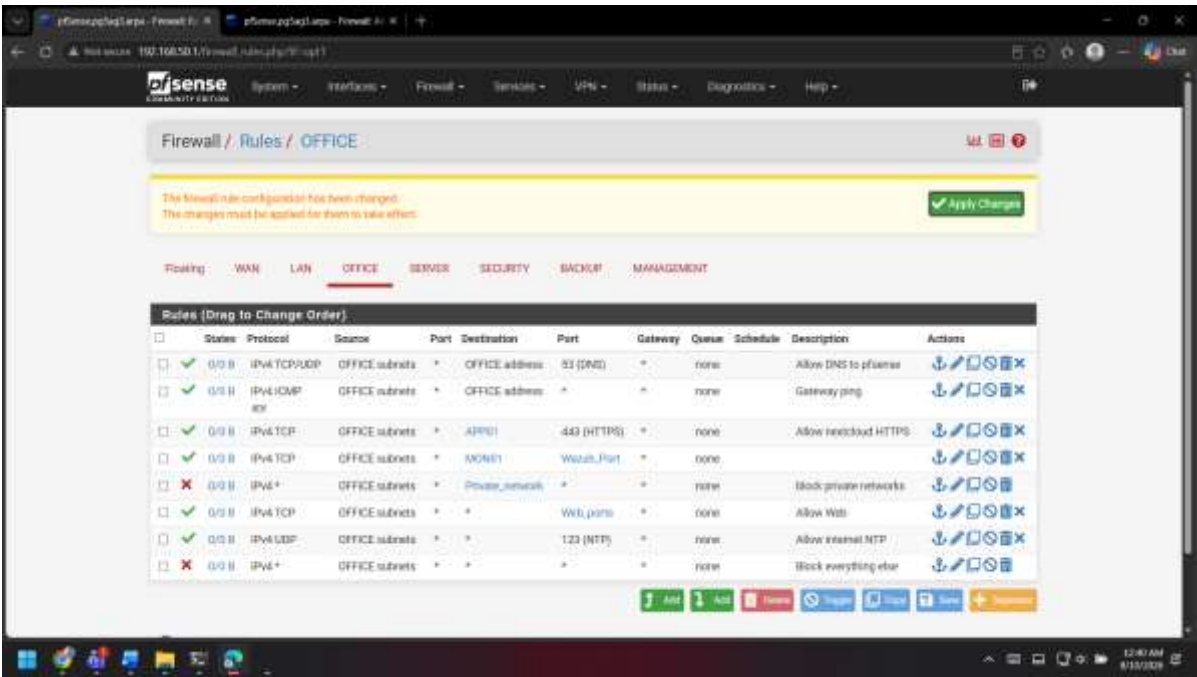


Figure B2. pfSense Office VLAN firewall rules used in the prototype.

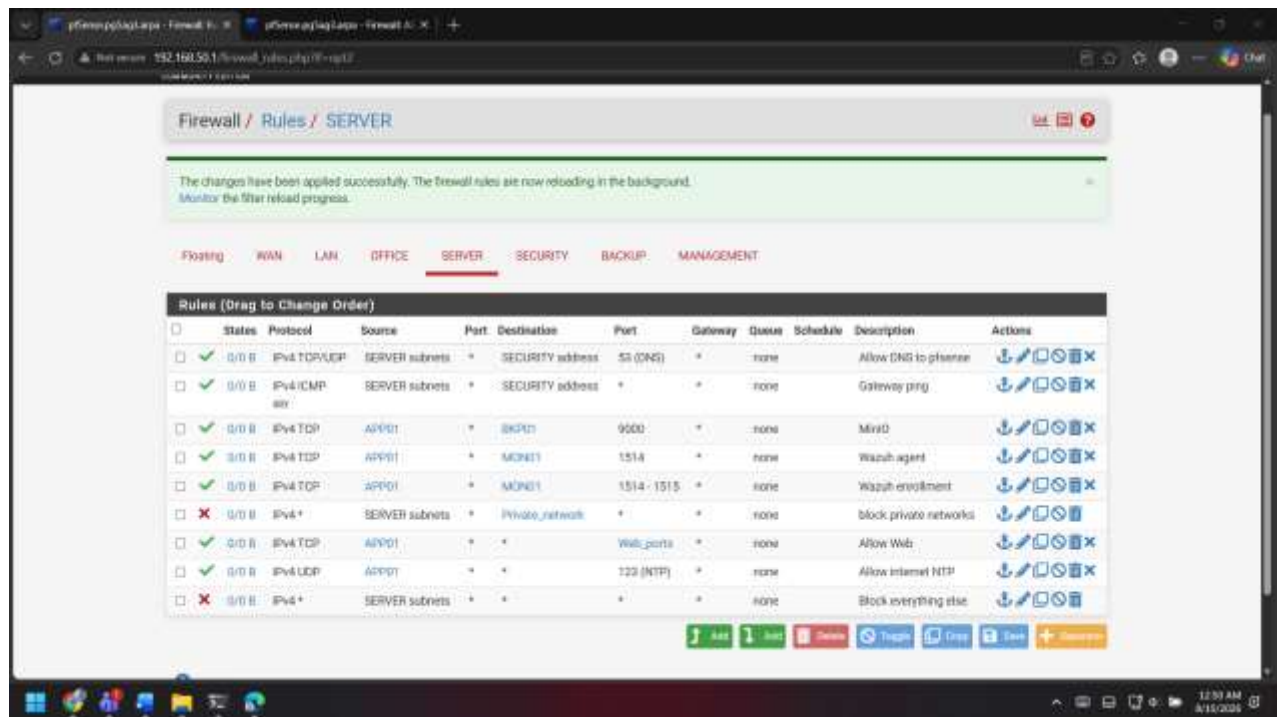


Figure B3. pfSense Server VLAN firewall rules showing the approved APP01 service paths.

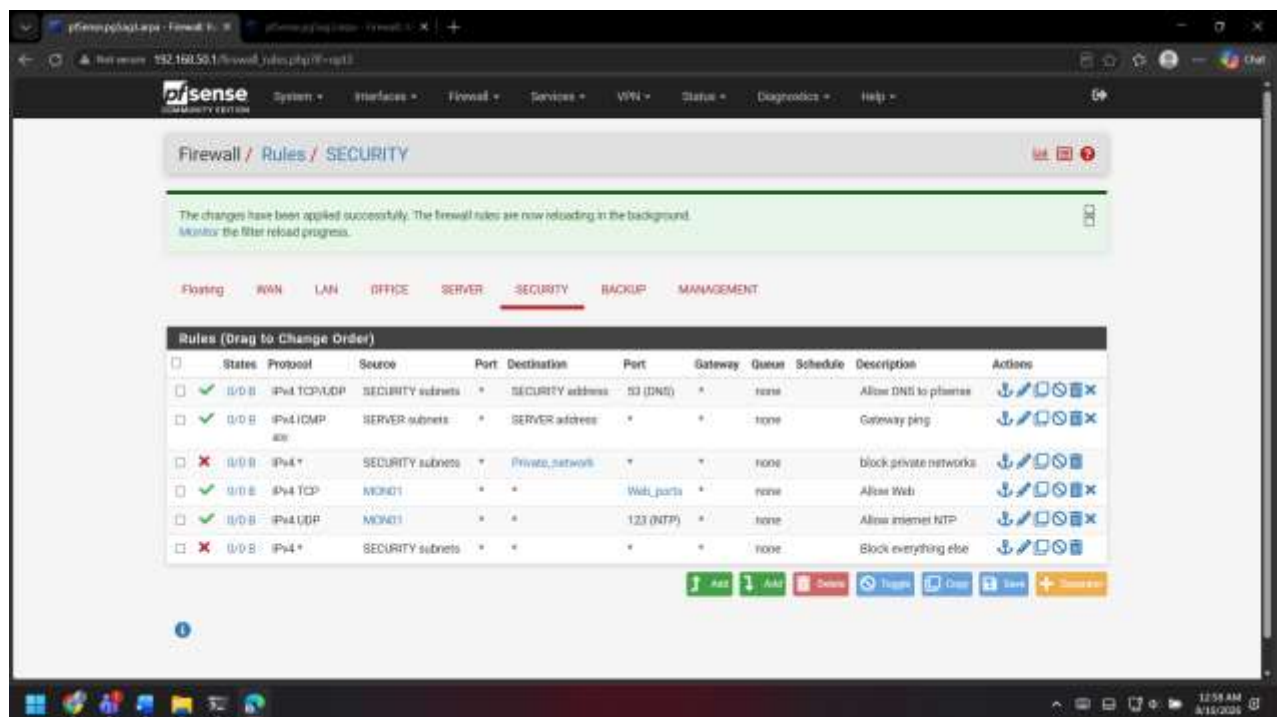


Figure B4. pfSense Security VLAN firewall rules used in the prototype.

Appendix C - Access Control Evidence

The screenshots below provide the application-level evidence for the executed access-control checks in Section 17.2.1. They show role-based folder visibility, successful Finance write access, denied cross-department access and the Nextcloud TOTP challenge. Evidence captured during prototype access-control verification on 29 September 2026.

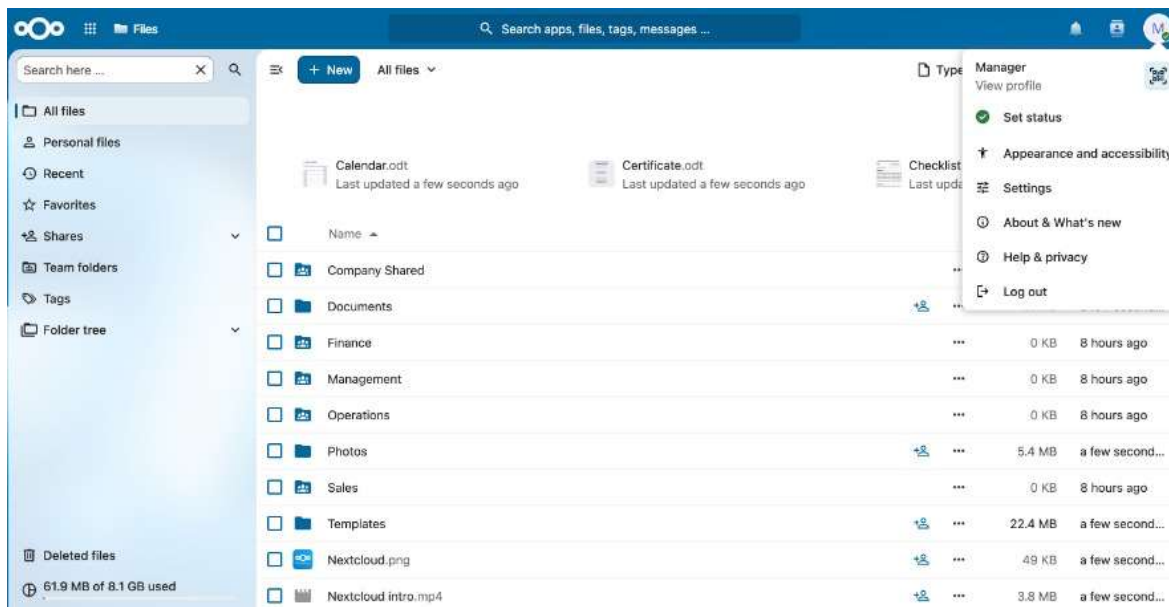


Figure C1. Manager account showing approved visibility of Company Shared, Finance, Management, Operations and Sales folders.

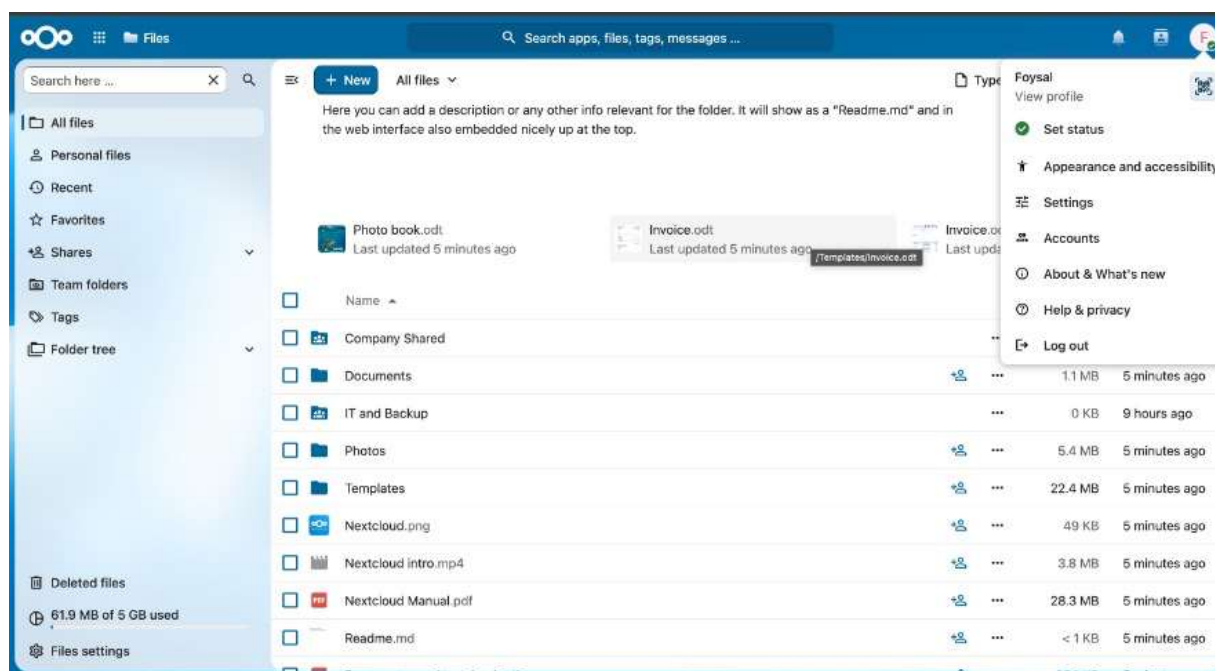


Figure C2. Foysal prototype account showing role-specific access to Company Shared and IT and Backup resources.

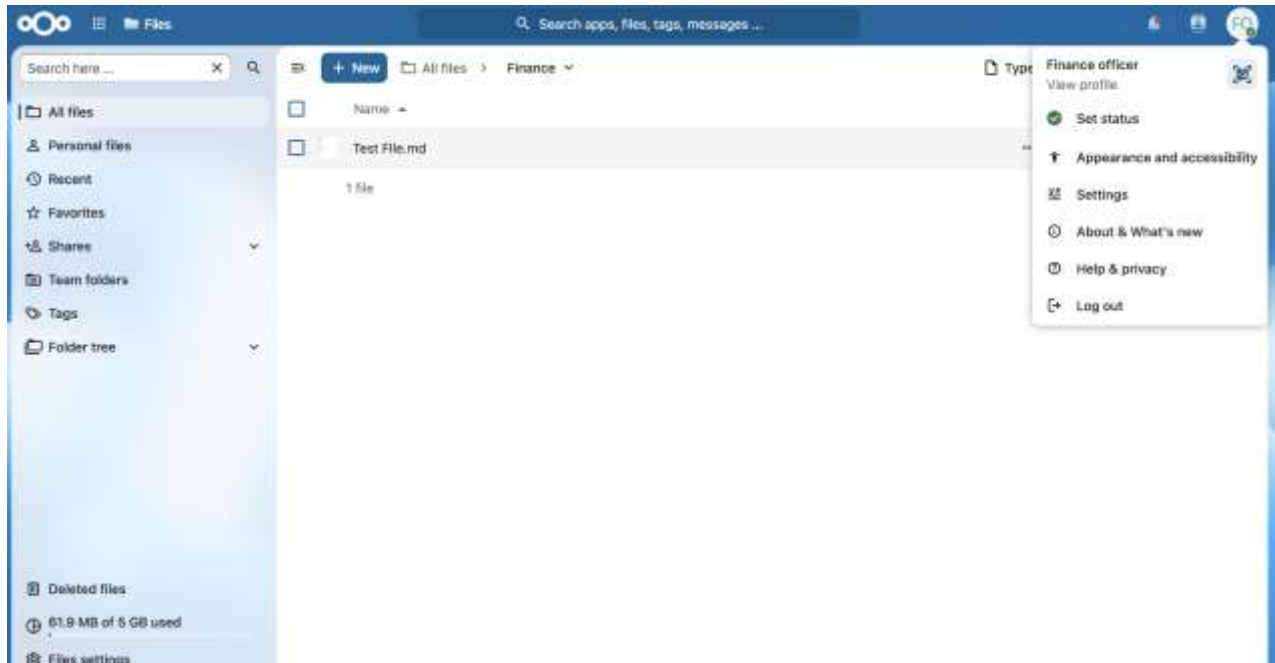


Figure C3. Finance officer inside the Finance folder with Test File.md, demonstrating successful Finance folder access and write capability.

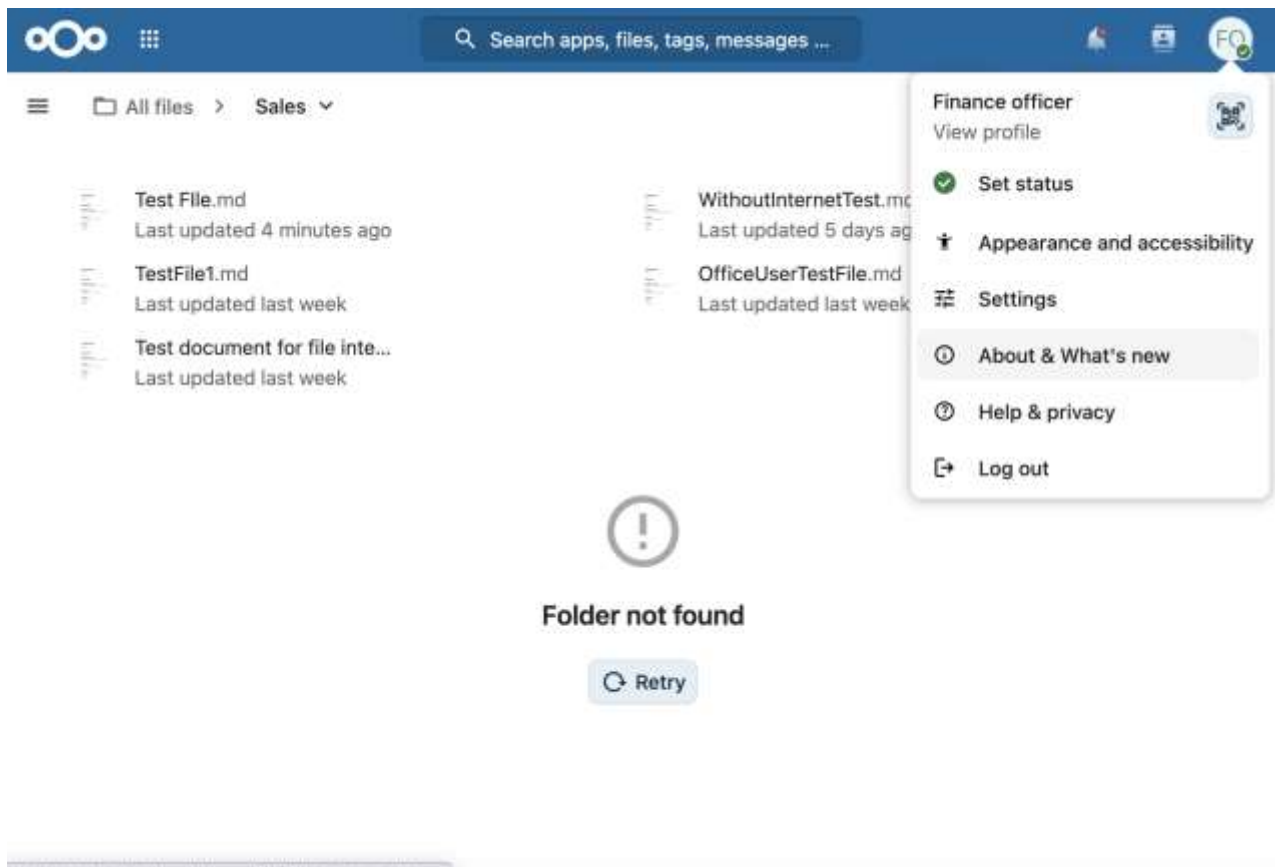


Figure C4. Finance officer direct attempt to open the Sales folder returning Folder not found, demonstrating cross-department access restriction.

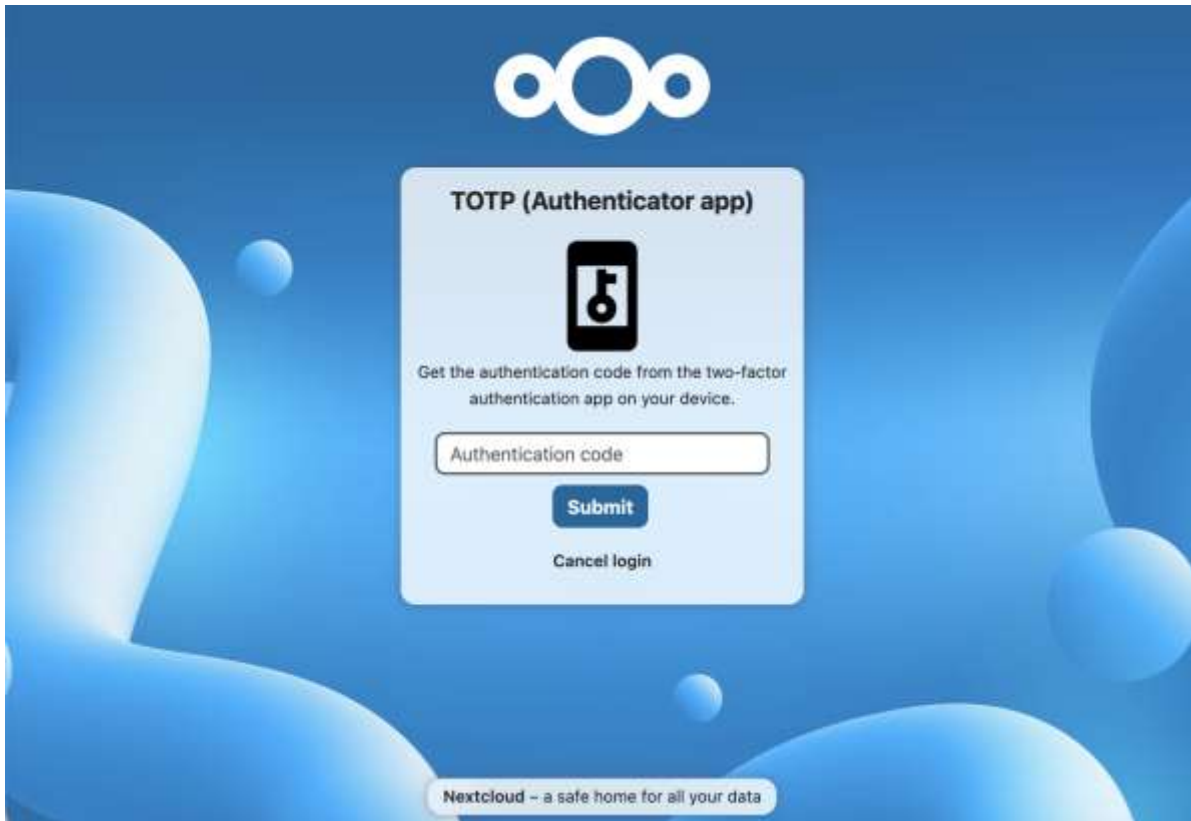


Figure C5. Nextcloud TOTP challenge displayed after administrator password authentication, demonstrating MFA enforcement on the prototype administrator login.